Trading Strategies for SPY ETF Based Options

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1 Strategy and Modelling

All 3 strategies adopted a consistent train/test split of 252/152 days, with the LSTM approach having additional validation split from the 252-day training set.

1.1 Iron Condor Approach

The core insight driving the strategy is that implied volatility in 0DTE SPY options tends to overstate the realized intraday move, especially after the initial open-hour turbulence subsides. By waiting until 10:30 AM and confirming that short-window realized volatility is declining, the strategy sells options that are rich relative to subsequent realized moves, collecting net premium that decays rapidly over the remaining trading session.

1.1.1 Strategy Implementation & Fine-Tuning

The strategy checks for entry exactly once per trading day at 10:30 AM. Three conditions must hold simultaneously before a position is opened:

- Volatility cooling: The 30-minute realized volatility (RV30) must be strictly less than the 60-minute realized volatility (RV60). Realized volatility is measured as the square root of the sum of squared log-returns, i.e., $RV = \sqrt{\sum r_i^2}$. A declining realized vol suggests the initial open-hour turbulence is fading.
- IV richness: The average Black-Scholes implied volatility of the two short legs must exceed the annualized 60-minute realized volatility: $IV_{short} > \underbrace{\text{IV_RV_ratio}}_{=1} \times RV_{60} \times \sqrt{390 \times \frac{252}{60}}$. This ensures the options being sold are priced above the recent historical volatility, giving a structural edge to the seller.
- Minimum net credit: The iron condor must collect a net credit at least \$0.02 per share (\$2.00 per contract), filtering out trades where the bid-ask spread would likely eliminate any edge. The net credit is defined as (short put price + short call price) - (long put price + long call price).

The short strikes are placed one intraday standard deviation from the current SPY price at 10:30 AM. The 60-minute realized volatility is scaled to the full remaining session (390 minutes) using a square-root-of-time rule, described by the following formula:

$$K_{sp} = S \times \left(1 - RV_{60} \times \sqrt{\frac{390}{60}}\right) \ , \ K_{sc} = S \times \left(1 + RV_{60} \times \sqrt{\frac{390}{60}}\right)$$

where K_p and K_c stands for the strikes for short put and short call respectively. The options whose strike is the closest to these calculated values are selected. For long positions, the strikes for call and put are calculated by first obtaining the median of the positive difference among all available strike before the entry time (if there's indeed an entry) as step size, then obtain the strikes as follows: (wing width is default set at 1, which is a multiplier to control the step size)

$$K_{lp} = K_{sp} - \text{wing width} \times \text{step size} \ , \ K_{lc} = K_{sc} + \text{wing width} \times \text{step size}$$

Again the options whose strike is the closest to these calculated values are selected. After entry, exit conditions are evaluated every 20 minutes until 3:00 PM. At each checkpoint, four exit conditions are evaluated in priority order:

- Take-profit: Close when unrealized P&L reaches 50% of initial net credit (premium short put + premium short call - premium long put - premium long call).
- Stop-loss: Close when unrealized loss reaches 1.5× the initial net credit.
- Short-strike breach: Close immediately if SPY crosses either short strike (K_{put} or K_{call}), reducing exposure to rapid directional moves.
- Time-based force exit: All positions are closed at 3:00 PM regardless of P&L.

The P&L is calculated via $100 *$ (entry credit $-$ exit debit), where the exit debit is defined by $P_{sp} + P_{sc} - (P_{lp} + P_{lc})$ (buy back shorting options, and sell longing options; notice you subtract this cost), all using closing price of the option under the same strike calculated above, and the constant 100 account for contract multiplier. Therefore, for options-specific considerations, as we can see from the previous P&L calculation, no bid-ask spread or transaction cost is considered. The implied volatility is calculated using Black-Scholes model, which is based on European option pricing, yet we're dealing with American options, therefore we're using Black-Scholes model as an approximation.

1.1.2 Backtesting & Performance Evaluation

The performance is provided by the Figure 1, 2, and Table 1. These results suggest that the strategy is quite selective. It trades infrequently because the entry filter is strict, but among the trades that do occur, the win rate is relatively high. That is consistent with the design of the code: it only enters when both the volatility and the IV condition are favourable.

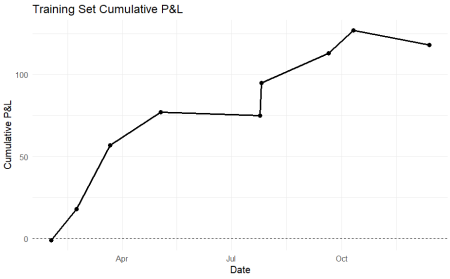


Figure 1: Cumulative P&L of Training Set (Iron Condor)

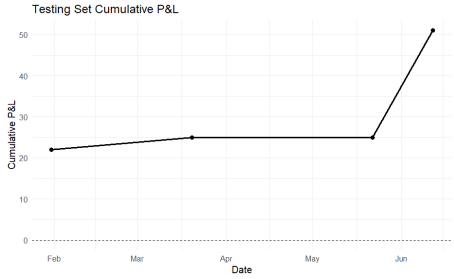


Figure 2: Cumulative P&L of Test Set (Iron Condor)

Metric	Value (Train)	Value (Test)
Total trades	9	4
Total P&L	\$118	\$51
Average P&L per trade	\$13.11	\$12.75
Median P&L per trade	\$18	\$12.5
Win rate	66.67%	75%

Table 1: Performance Summary (Iron Condor)

1.2 Shorting Straddle Approach

This strategy sells an at-the-money 0DTE SPY straddle on days when intraday volatility looks like it's calming down. The idea is that when short-term vol has been dropping relative to the broader morning trend, the market probably isn't going to make a big move - so the premium priced into the straddle ends up being more than what actually gets realized. Rather than selling every day, we only enter when two specific vol conditions are met, which filters out most of the days where this trade tends to lose.

Compared to the baseline in Objective 1, which buys a straddle on high-vol days and holds to expiry, this is the opposite bet. The baseline profits from large moves; this one profits when the market stays quiet. The strategy is largely the same as what was described in the proposal - the entry logic, exit rules, and risk controls were all in place. The main change since then is the addition of the IV proxy as a supplementary signal.

1.2.1 Strategy Implementation & Fine-Tuning

At 11:00 AM each day, we check two conditions. If both are satisfied, we enter at 11:01 AM:

- $\sigma_{15} < 0.9 \times \sigma_{60}$: the 15-minute realized vol is less than 80% of the 60-minute realized vol. This means short-term price movement has slowed noticeably relative to the morning.
- $\sigma_{60} <$ 20-day rolling median of σ_{60}: overall vol on this day is below its recent baseline, filtering out days where the market is just broadly elevated.

When both conditions hold, we sell one ATM call and one ATM put at the strike closest to the 11:00 AM SPY price. Entry prices use the option close prices at 11:01 AM with \$0.02/leg slippage applied on both sides. We use 11:00 AM as the check time because the market has usually settled from the open by then, so the vol readings are more reliable. A fixed entry time also avoids any look-ahead bias.

Three exit conditions are checked every minute after entry:

- Stop-loss: if $|S - K| > 0.8\times$ premium, close immediately. This keeps the max loss on any trade roughly in line with the premium collected.
- Vol expansion: if $\sigma_{15} > 1.50 \times \sigma_{60}$ after entry, close. This is the entry signal flipping - if vol suddenly spikes post-entry, the regime assumption is broken.
- Time exit: close at 14:30 no matter what. Gamma gets large enough in the last 90 minutes of a 0DTE option that holding through it isn't worth the risk.

If none of these trigger, the position settles at expiry using the 15:59 SPY close price with the standard call/put payoff formulas.

For parameter set up, please refer to 5. As an additional signal, we compute an IV proxy defined as the straddle price divided by the SPY reference price, which approximates twice the implied volatility scaled by time. Days where this proxy exceeds 110% of its 30-day rolling median are flagged as 'rich', indicating that the options market is pricing in more movement than typical - favourable conditions for selling premium.

For options-specific considerations, since we're trading ATM straddles on SPY, liquidity is good and the bid-ask is narrow, so the \$0.02/leg slippage estimate is conservative. The 100x multiplier is applied throughout the P&L calculation. SPY options are American-style, so there's technically early exercise risk on the short call, but same-day expiry makes this fairly unlikely in practice and the 14:30 time exit reduces it further.

1.2.2 Backtesting & Performance Evaluation

The strategy was evaluated on the first 252 trading days (2023-01-03 to 2024-01-03) as the in-sample training period, with the remaining days (2024-01-04 to 2024-06-28) used as the out-of-sample test period.

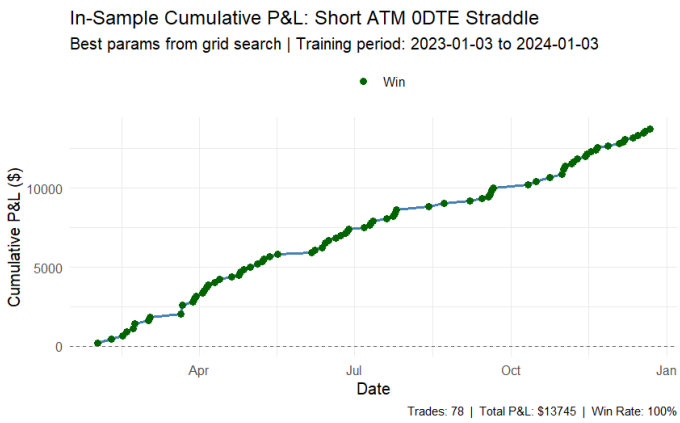


Figure 3: Cumulative P&L of Training Set (Shorting Straddle)

Date	Strike	Entry	Exit	P&L	Exit Reason
2023-01-31	403	11:01	14:30	202.55	TIME_EXIT
2023-02-08	412	11:01	14:30	237.23	TIME_EXIT
2023-02-15	411	11:01	11:48	210.02	STOP_LOSS
2023-02-17	405	11:01	14:30	253.66	TIME_EXIT
2023-02-21	401	11:01	14:17	235.02	STOP_LOSS
2023-02-22	399	11:01	14:03	276.35	VOL_EXPANSI
2023-03-02	394	11:01	13:35	220.30	VOL_EXPANSI
2023-03-03	401	11:01	12:54	200.11	STOP_LOSS
2023-03-21	396	11:01	14:30	217.11	TIME_EXIT
2023-03-22	399	11:01	14:00	534.84	VOL_EXPANSI

Table 2: Trading Summary (first 10 trades) (Shorting Straddle; Training Set)

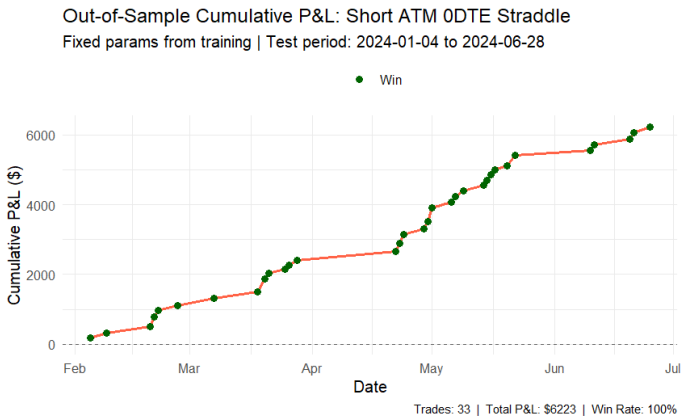


Figure 4: Cumulative P&L of Test Set (Shorting Straddle)

Metric	In-sample (Training)	Out-of-Sample (Testing)
Number of trades	78	33
P&L	$13,745	$6,223
Average P&L per Trade	$176.22	$188.57
Win Rate	100%	100%
Payoff Ratio	N/A	N/A
Max Drawdown	$0	$0

Table 3: Performance Summary (Shorting Straddle)

The win rate of 100% and zero drawdown are unusually strong and warrant caution. The N/A payoff ratio is exactly due to this fact, since there're no losing trades. The stricter VOL_COMPRESS_MULT of 0.90 selected by the grid search filters entry to only the calmest days in the training period, resulting in fewer but consistently profitable trades. The out-of-sample period (2024-01-04 to 2024-06-28) produced 33 trades with a total P&L of $6,223, average P&L of $188.57, and a Sharpe of 45.88, which is consistent with training results.

1.3 LSTM Approach

This approach is adapted from Tan et al. (2024), which used daily data and monthly straddle (hold the same straddle until the month end), thus they only have to calculate the entry signal, since they're merely adjusting the position size (rebalancing) at each time step; while. The difference is that we're using minute level data, and since we're not holding monthly options (American options can be early exercised), we're calculating the entry signal for each time window (see below fo details), and hold the position until the next entry, where we'll calculate a different set of straddle with different strike price.

An assumption is that the price information of the straddle is continuous given a specific time. More specifically, for ETF/derivatives whose trading detail is missing for the last minute of the time window, we assume that the price is staled from the closest time point. The time window is set as 15 minutes, starting from the beginning of the day (9:30 am), and use the first minute as the timestamp of this time window. For example, 9:30 am represent the data from 9:30 am to 9:44 am. Each window collapse the data in between and only keep the open price of the first minute and closing price of the last minute.

The assumption and data aggregation technique mentioned above is not an issue for SPY ETF, because for assets that are traded in minute level, the closing price of the current minute is the open price of the next minute, however for options since the derivatives are not traded in minute level, we need to impose this extra assumption. For example, for the time window from 9:30 am to 9:44 am, if a call option given strike K only have data from 9:30 am to 9:40 am, we'll assume that the price is stale from 9:40 am to 9:44 am, and the closing price of this 15-minute time window will be the closing price at 9:40 am. This 15-minute window is a hyperparameter that is currently being fixed (not optimized by Optuna), and will be changing in the ensemble method proposed in the future.

There are 3 components of this LSTM approach, including feature construction, objective function, and hyperparameter search, which we will describe in the following subsections. For the technical details on the LSTM structure, please refer to section A in the appendix.

1.3.1 Feature Constructions

All the features below are constructed to be straddle specific. We start by selecting an appropriate strike for the straddle based on moneyness (see feature 4, core option features), and the features are restricted to straddle of this specific strike price. Some of the features reset calculation at each day, meaning that relevent calculation (for example volatility) only uses data in the same day, while some features uses cross day data, yet this does not mean that we hold a staddle overnight. We're merely calculating, for example, the standard deviation by using the return of the straddle at this strike, at each bar. For each 15-minute bar with a call and put option available at this strike, it will be able to form a hypothetical straddle and we're using the closing and open price of this straddle to calculate the log return. By collecting these returns within each bar, we're able to calculate the rolling window summary statistics. For features that involves data from different days, it allows the LSTM to remember more historical states, while features only involving current day data focus more on daily price movement.

Daily Volatility Estimate σ_t We first cover the daily volatility estimate σ_t used in the construction of the first feature. This estimate only calculate the Exponential Weighted Moving standard deviation (EWMSD) of the straddle log returns within each day, therefore the data used to calculated σ_t does not involves data in day $t-1$. Any bar where the within-day EWMSD has not yet stabilised (e.g. the shifted first bar of each day is NaN because EWMSD requires ≥ 2 observations) is replaced by the expanding **median** of all valid σ^{raw} values seen so far across the full dataset. If the expanding median is itself NaN (i.e. no valid EWMSD value has been produced yet, as at the very start of the dataset), the hard fallback $\sigma_t = 10^{-4}$ applies. The backtest further caps the resulting volatility-scaling $\left(\frac{\sigma_{target}}{\sigma_t}\right)$[1] at 20.0. The span is chosen to be 26 (the whole day) to assign exponential weights to closer time points, and the weights are calculated recursively.

Eventually, we define the following input feature. Notice that these features are adapted from Tan et al. (2024):

1. Normalized Return: defined as $\frac{r_{t-k}^{straddle}}{\sigma_t^{straddle}\sqrt{k}}$, which representing straddle return from time $t-k$ to t, normalized by daily volatility estimates (mentioned above) scaled to a time scale k. This k is readjusted to align with our minute level data, with value $k \in \{1, 4, 8, 13, 26\}$, where the values represent the numbers of time windows, in our case, 15 minutes. Given our data, which range from 9:30 am to 15:59 pm, $k = 26$ represents daily. All numeric values without specific mentioning follows this rule, i.e., number of units in terms of 15-minute time window. The value of this normalized return is clipped at $[-10, 10]$ to prevent extreme values from dominating in case σ_t is small.

2. MACD Indicators: defined by the volatility normalized moving average convergence divergence (MACD) indicators (Baz et al., 2015). More specifically:

$$
\begin{aligned}
MACD(i,t,S,L) &= EWMA(i,t,S) - EWMA(i,t,L) \\
MACD_{norm}(i,t,S,L) &= \frac{MACD(i,t,S,L)}{std\left(\text{stradle prices}_{t-5:t}\right)} \\
&\quad \text{(expanding fallback applies for std } (< 2 \text{ data points), clip lower bound } 10^{-6}) \\
Y_t^{straddle} &= \frac{MACD_{norm}(i,t,S,L)}{std\left(MACD_{norm}(i,t-20:t,S,L)\right)} \qquad (\text{ clip at } [-3,3]) \\
X_t^{straddle} &= \frac{1}{3}\sum_{k=1}^{3}\gamma\left(Y_t^{straddle}(S_k,L_k)\right) \;\;,\;\; \gamma(y) = \frac{1}{0.89}y\exp\left(-\frac{y^2}{4}\right)
\end{aligned}
$$

   where $S_k \in \{2, 4, 8\}$, $L_k = \{8, 16, 32\}$, in total 3 ordered pairs, and EWMA represents exponential weighted moving average. The time scale $j = S_k$, L_k corresponds to the half life parameter $HL = \frac{\log(0.5)}{\log\left(1-\frac{1}{j}\right)}$ used in EWMA.

3. Option Momentum Features: defined as the average return of straddle over look back period of n numbers of 15-minute window, specifically, $r_{t-n:t} = \frac{1}{n}\sum_{i=t-n}^{t-1} r_i$, where $n \in \{1, 5, 10, 20\}$ (therefore this contribute 4 features in total). This rolling window does not reset at each day, therefore it will include return from previous day for large n.

4. Core Option Features: log-moneyness and DTE (days to expiry, in years). Each of the concept is defined as follows:

   - moneyness: for call option, moneyness $= \frac{S}{K}$, and for put moneyness is defined as moneyness $= \frac{K}{S}$. Notice when selecting call and put options, since it's usually not possible to select optionswith moneyness exactly equal to 1, we select the pair of options that are closest to at-the money within a moneyness range of 0.95 to 1.05. We take the log of this constant to form the value for log-moneyness.

     This moneyness and strike optimization is computated prior to fitting the model, therefore we're not calculating the set of call and put option when we detect a significant signal ($|X_t| \geq \theta$; see below for details), but instead these values are precomputed. However, note that for a given time window t, it's using open price of time t, and closing price of $t + 15\,\text{min}$, therefore for this selection we shouls use the open price of SPY ETF, because this price is exactly known at time t, to prevent future leakage.

   - DTE: there're 390 trading minutes per day (9:30 am to 15:59 pm), and there're 252 trading days annually, thus DTE is defined as the following:

$$
\frac{minutes(15:59 - \text{time of trade})_{day\,t}}{252 * 390}
$$

[1] See section 1.4 for details.

5. Rolling Volatility[2]: we consider the rolling volatility of the log-return of SPY ETF, i.e., $\log\left(\frac{R_{close,\,t}}{R_{open,\,t}}\right)$, with rolling window begin 5, 10, and 20, starting from $t-1$. Notice that this feature does not reset each day, unlike what we did for feature 1, therefore this rolling volatility calculation involves cross-day volatility calculation. This new feature is introduced to provide the model with certain prediction power on the volatility.

The major input structure of LSTM consists of 2 parts: input sequence, and input vector at time t. The input sequence is a collection of input vectors, and each time only 1 input vector will be used as an input. The input vector is a stacked feature vector over the past 10 (denoted by τ)[3] windows, i.e., to calculate the input vector of time t (used to generate the entry signal of the trading), we use data from $t-10$ to $t-1$ (to avoid future leakage), and each time (window) we could calculate a corresponding feature vector $\left(\in \mathbb{R}^{19}\right)$, and then we stack these vectors to form a vector $\left(\in \mathbb{R}^{190}\right)$. This is a shifting window approach, since for a given time point t, we consider the past 20 time window.

The output feature is the entry signal for each time window t, denoted as $X_t \in [-1, 1]$. This range makes tanh the perfect activation functions for the output layer (see section A for details), and what it represents is the size of the position, along with whether we're longing or shorting the straddle. To reduce high frequency trading cost and insignificant trading, we set the significant threshold for the trading signal to be $|X_t| \geq \theta$, where θ is a parameter being optimized, initially set as 0.35 to avoid high-frequency trading.

Everytime there's a new entry signal detected or until the end of the day, whichever comes earlier, we'll close the previous position, and reopen the position using the new set of the strike and options calculated in the moneyness procedure, therefore this strategy account for both flexible entry and exit signal.

1.4 Objective Function

We take the risk of the portfolio in account when concstructing the objective function, compared to Tan et al. (2024), which only uses Sharpe-Ratio. The objective function is given as follows:

$$E(\cdot) = -\frac{\frac{1}{|\Omega|}\sum_{\Omega}\tilde{r}(t)\times\sqrt{252}}{\sqrt{\frac{1}{|\Omega|}\sum_{\Omega}\tilde{r}(t)^2 - \left[\frac{1}{|\Omega|}\sum_{\Omega}\tilde{r}(t)\right]^2}} + \lambda_1 \cdot ES_\alpha - \lambda_2\left(\frac{1}{|\Omega|}\sum_{\Omega}\tilde{r}(t)\times\sqrt{252}\right) \quad ,$$

$$\tilde{r}(t) = \tilde{r}_{t,t+1} = R(t) - c\cdot\sigma_{target}\left|\frac{X_t^{straddle}}{\sigma_t} - \frac{X_{t-1}^{straddle}}{\sigma_{t-1}}\right| \quad , \quad R(t) = X_t^{straddle}\left(\frac{\sigma_{target}}{\sigma_t}\right) r_{t,t+1}^{straddle}$$

where $\Omega = \{t \,|\, t = 1, \ldots, T\}$ represents collection of all trading time, and c is a measure of average transaction cost in basis points. σ_{target} is the target volatility of the portfolio, which is a user specified constant, currently set as 15% (Tan et al., 2024). $\tilde{r}(t)$ is know as the turnover-adjusted return, used to account for the position change between 2 consecutive trades, thus the consideration of transaction cost is incorporated in this objective function. ES_α is the expected shortfall, with $\alpha = 5\%$. Since we only have 1 possible path of the prices, to calculate this expected shortfall we naively collect all the returns of all trading points and calculate the mean of the largest $\alpha = 5\%$ losses. The multiplier λ_1 and λ_2 controls the level of penalty on the risk and the return of the portfolio. The involvement of the mean return is to force the model to not only considered risk-adjusted return (since if the return has volatility close to 0, but slighly larger mean return, it will generate a large risk-adjusted return, but the actual profit could be miniscule), but also aiming at optimizing the P&L, as we'll see in B that the .

Table 5 provided detailed hyperparameter values used for this machine learning approach. The transaction cost in the P&L calculation is computed by simply using $0.02 * 2 * (X_t - X_{t-1}) * 100$, where 100 is the option multiplier, and 0.02 is the bid-ask spread we choose for the call/put option, and the constant 2 is to account for both options. For P&L calculation, in order to maintain a consistent risk profile, a volatility scaling is applied: $VS_t = \min\left(\frac{\sigma_{tgt}}{\max(\sigma_t, 10^{-4})}, 20.0\right)$, where the σ_t is the realized volatility over the period that we hold the current straddle; the 20 bound is to prevent position size to blow up if the volatility is close to 0. The σ_{tag} is again the target volatility of the portfolio, speficied as 15% by Tan et al. (2024).

The P&L calculation is provide as Gross $\text{P\&L}_{t+1} = X_t \times VS_{t+1} \times (Close_{t+1} - Close_t) \times 100$ if holding straddle of the same strike for certain period; however, if we're switching to a straddle with different strike, the return within this switching is determined by Gross $\text{P\&L}_{t+1} = X_t \times VS_{t+1} \times (Close_{t+1} - Open_{t+1}) \times 100$. For both cases, we consider transaction cost and the bid-ask spread by incorporating the 0.02 value mentioned above.

1.5 Performance Evaluation

We conduct train/test split by keeping the first 252 trading days as training set, and the remaining 152 trading days is used as the test set. The validation set is the last 10% of the training set. Numerical sesults are shown in Table 4, while the visualized result including path of cumulative P&L and daily signals are presented in section B in the appendix.

[2]This is a new feature added to the LSTM approach, therefore only realized volatility is used, as this feature hasn't been tested to be effective, especially in the training set; see section B for details on model evaluation; we'll consider implied volatility after the model has been proved to have good performance on both training and testing dataset.

[3]This is a hyperparameter that is optimized; for further discussion on hyperparameter search, see section C in the appendix

Data	Total Net P&L	Win Rate	Payoff Ratio	Maximum Drawdown	Number of Trades	Position Frequency		
						Long	Short	Flat ($X_t = 0$)
Train	\$-1983.33	44.4%	1.063	\$4060.37	1333	0.1%	62.0%	37.9%
Validation	\$81.17	50.0%	1.240	\$568.05	68	0.0%	36.2%	63.8%
Test	\$97.44	56.6%	1.060	\$944.02	694	0.0%	61.5%	38.5%

Table 4: Performance Summary (LSTM)

We could see that the model did a bad performance on the training set, and overall better performance on the validation and test set in terms of the terminal P&L and win rate, indicating less likelihood of overfitting. The major component of the trading is consists of shorting positions, with little long positions only appear in the training stage.

2 Summary, Conclusion and Further Study

For the strategy involving iron condor, the model uses Black-Scholes to estimate implied volatility, even though SPY options are American. This approximation can differ especially for put options. Second, checking entry only at 10:30 AM and exits only every 30 minutes may miss better opportunities or faster-moving stop-loss events between checkpoints. In addition, while parameters were chosen on economic grounds, the parameter choices are not fully optimized and validated out-of-sample. Finally, it use close price and didn't include bid ask spread, since there are four legs, the transaction costs will lower the return. Possible improvements will include using pricing model spefically designed for American options, incorporating transaction cost, bid-ask spread, and using more advanced models to predict intraday volatility or breakout risk.

In terms of shorting straddle approach, the regime filter does seem to be adding value. By only trading on days where vol is compressing both short-term and relative to the recent baseline, the strategy avoids high-vol days where short straddles typically lose. Parameters were selected via grid search on the training data, which addresses the data snooping concern raised in the draft. That said, the 100% win rate on training is likely a consequence of overfitting to a favourable low-volatility regime in 2023 - the filter is strict enough that it only captures the easiest trades. The out-of-sample results are consistent but cover only 6 months, so broader testing would be needed to confirm robustness. The strong results may also partly reflect a favourable macro environment for short-vol strategies in 2023-2024. Potential future improvement includes the following: first, the entry only checks at 11:00 AM. Scanning every few minutes from 10:00 to 1:00 would probably find more and better entry points, though we would like to have a one-trade-per-day cap to avoid overtrading. Second, the 14:30 time exit ignores P&L entirely. If the trade is already well in profit near close, it's probably better to lock it in rather than hold to the hard cutoff. A time-weighted profit target would handle this better. Current value are taken as a conservative step to avoid data snooping, yet no hyperparameters were optimized on the training data, therefore existing room to improve performance through a proper walk-forward grid search over the key thresholds.

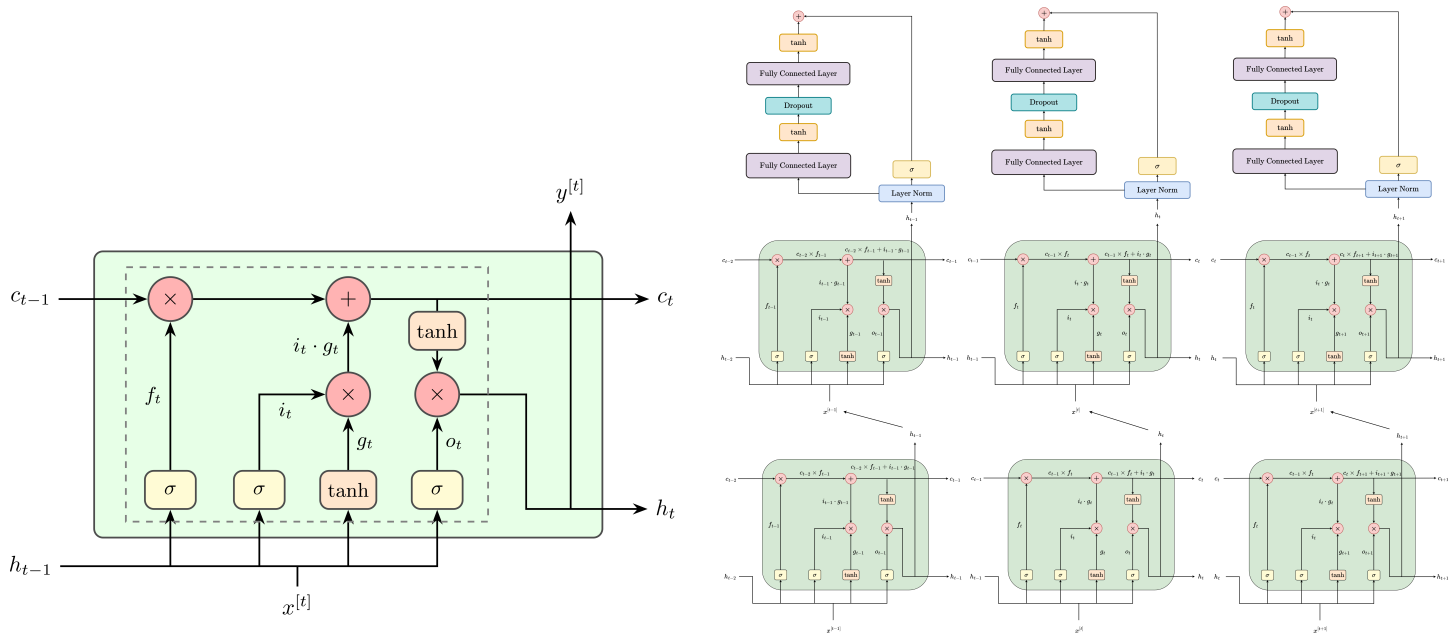
The LSTM approach does not appear to overfit to the training data, yet despite the postive P&L of the model on the validation and the test set, the model didn't perform well on the training set, ending up a negative P&L. The next step is to focus on emsemble method (Kryńska and Ślepaczuk, 2022), by trying window lengths other than 15 minutes. If time permits, hybrid models (for example, combining ARIMA/GARCH with LSTM) will also be tested, particularly given that the model did not provide a decent performance on the training data, incoporte GARCH model into the rolling volatility feature (having the model redict the residual after considering GARCH forecast) could provide promising result.

A Technical Details of Long Short-Term Memory

Figure 5a illustrate the structure of a signle LSTM block. There are 4 stages in total, namely forget, input, cell, and output, along with relevent operation. Let $x^{[t]} \in \mathbb{R}^{d_x}$, and `hidden_size` $= d_h$, then the cell state vector c_t, hidden state vector h_t both $\in \mathbb{R}^{d_h}$. We have the following:

$$\begin{cases} \text{Forget gate} : f_t & = \sigma\Big(\underbrace{W_{if}}_{d_h \times d_x} \underbrace{x^{[t]}}_{d_x \times 1} + \underbrace{b_{if}}_{d_h \times 1} + \underbrace{W_{hf}}_{d_h \times d_h} \underbrace{h_{t-1}}_{d_h \times 1} + \underbrace{b_{hf}}_{d_h \times 1}\Big) \\ \text{Input gate} : i_t & = \sigma\left(W_{ii}\, x^{[t]} + b_{ii} + W_{hi}\, h_{t-1} + b_{hi}\right) \\ \qquad\qquad g_t & = \tanh\left(W_{ig}\, x^{[t]} + b_{ig} + W_{hg}\, h_{t-1} + b_{hg}\right) \\ \text{Output gate} : o_t & = \sigma\left(W_{io}\, x^{[t]} + b_{io} + W_{ho}\, h_{t-1} + b_{ho}\right) \\ \text{Cell gate} : c_t & = c_{t-1} \odot f_t + i_t \odot g_t \\ \text{Hidden state} : h_t & = o_t \odot \tanh(c_t) \end{cases}$$

Notice that all weight matrices shares the same dimension, and similarly for the bias vector. The operations elaborate the operation happens inside the LSTM block, and each block's output is used as the next block's input. The cell state c_t represents the long-term memory, while the input $x^{[t]}$ represens the short-term memory.



(a) LSTM Block

(b) Full Model Example (2 LSTM layers)

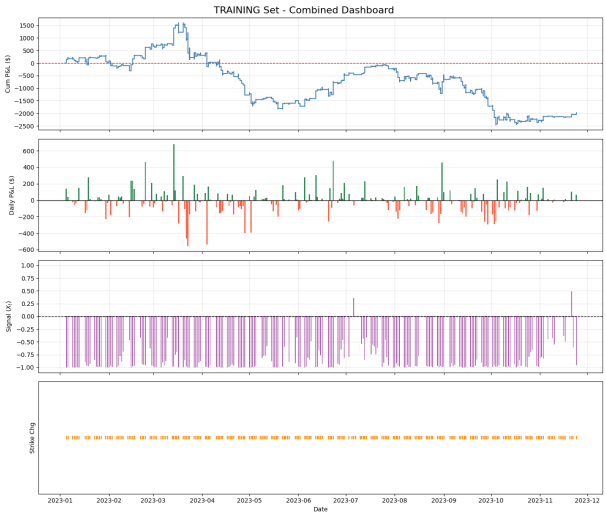
Figure 5: Illustration of LSTM and Model Structure

Figure b demonstrates an example of a 2 layer LSTM, merely stacking another layer of LSTM above the output. In the current set up, the model is a many-to-many relationship, since we have multiple input as a sequence, and each input generate 1 output (entry signal). There're also other relationships such as 1-to-1, many-to-1 (for example, you only want to predict the next character), or 1-to-many, which we will not go into details.

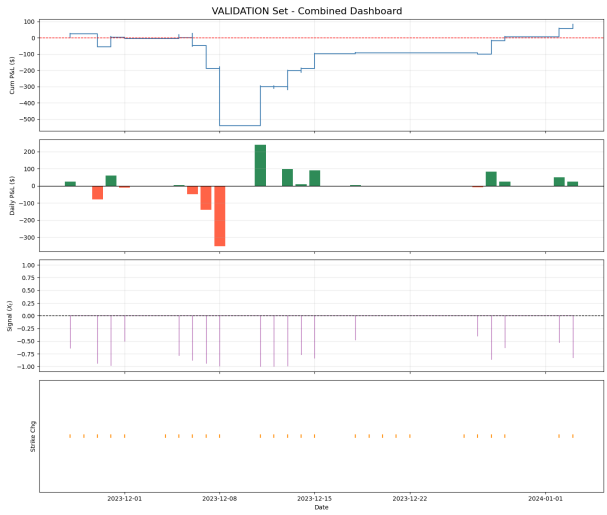
The output $X_t \in [-1, 1]$, therefore the activation function for the output layer is tanh. Before we talk about the final output, the output from each LSTM block is first fed into a fully connected (FC) linear layer, whose number of nodes equal to d_h, i.e., equals to the hidden dimension of the LSTM. The activation function for this hidden layer is again tanh. The second FC layer only contains 1 node, therefore aligns with the single output X_t after taking tanh. A drop out layer is added before the second FC layer and after the activation function for the first FC layer (tanh), but is omitted from the graph for brevity. This structure applies to each LSTM block, thus having 1 signal for each time-point. We adopted a residual connection that add the output after the second FC layer and the initial hidden output (after layernorm and sigmoid; both are used for numerical stability reason, but the sigmoid is only applied for the residual connection, while layernorm output is used by all FC layers) to improve the model performance. The sigmoid is applied to the hidden output because we want to increase the tendency of long position, as we could see in the next section, majority of the trades are taking short positions of thr straddle, making them only have a limited space for profitability, thus highly risky. This approach demonstrates better performance in the test and validation set, compared to the training set.

B Performance Evaluation - LSTM

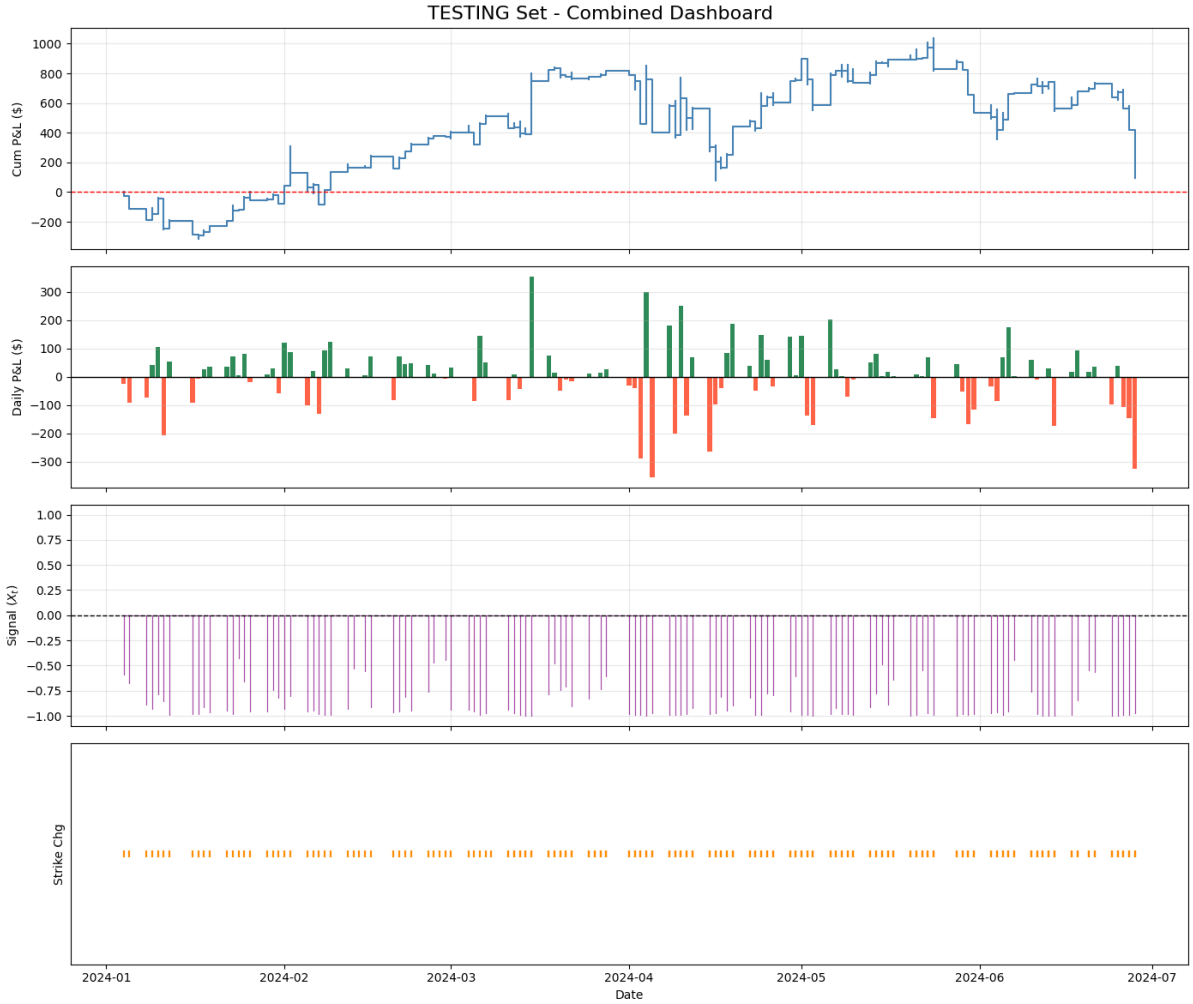
We provide the cumulative P&L, daily P&L, signal size, and strike change for training, validation and test set. The reader may cross reference Table 4 for numerical details.



(a) Training Set



(b) Validation Set



(c) Test Set

Figure 6: Performance Summary of LSTM Approach

C Hyperparameter Settings

- Iron Condor Approach: Parameters were chosen on economic grounds rather than grid search. With only 13 trades, data-driven optimization would be severely overfit. The 50% take-profit is a standard short-premium heuristic, not fitted to the current sample.
- Shorting straddle Approach: all parameters were chosen ad hoc before running the backtest; no optimization was performed on the training data.
- LSTM Approach: These hyperparameters are optimized using Optuna, which effciently prunes experiments with exploding gradient or NaN results. The number of epoch of hyperparameter search is 100 per experiment, which is sufficient compared to the 300 epochs for the actual run, since we have incorporate early stopping patience (20 epochs; depending on whether the validation loss is improving). The number of experiements is 500 trials.

Hyperparameter	Value	Search Grid	Remark
Entry Time	10:30 AM	[10:30, 11:00]	
Wing Width	1 strike step	[1.0, 1.1]	Used to detemine strike of long position
Min Net Credit	0.02×100		One of entry criteria; filters near-zero condors that would not survive friction costs (bid-ask spread)
IV_RV_ratio	1		
Take Profit	50%	[1.0, 1.1]	First exit crteria
Stop-Loss	1.5	[1.5, 2.0]	Second exit crteria
Force Exit	3:00 PM		Fourth exit crteria
Exit Check Freq	Every 20 min	[20, 30]	

(a) Iron Condor

Hyperparameter	Value	Remark & Search Grid
`CHECK_TIME`	11:00 AM	$\sim$ 90 min after open; market likely stabilized [10:30, 11:00, 11:30]
`VOL_COMPRESS_MULT`	0.90	σ_{15} must be below σ_{60} to confirm compression [0.70, 0.80, 0.90]
`VOL_LOOKBACK_DAYS`	20	Roughly one trading month of context [20, 30]
`STOP_MULT`	0.80	Closes position before losses exceed the collected premium [0.80, 0.90]
`TIME_EXIT`	14:30	Gamma risk (options) increases sharply in the final 90 min [14:15, 14:30, 14:45]
`VOL_EXPAND_MULT`	1.5	Clear reversal signal when short-term vol surges [1.5, 2.0]
`SLIPPAGE_PER_LEG`	$0.02	Reasonable estimate for ATM SPY options bid-ask

(b) Shorting Straddle

Hyperparameter	Value	Remark
`hidden_size`	40	Search Grid: [40, 80, 160]
`num_layers`	1	Search Grid: [1, 2, 3]
Drop Out	0.5	Search Grid: [0.1, 0.5, step=0.1]
Learning Rate	0.0003[4]	Search Grid: [1e-5 to 1e-2]
Lookback Window (τ)	10	Search Grid: [10, 60, step = 10]
λ_1 (ES factor)	0.4	Search Grid: [0.1, 0.5, step=0.1]
λ_2 (mean return factor)	0.2	Search Grid: [0.04, 0.2, step=0.02]
Minibatch Size		Search Grid: [32, 64, 128, 256]

(c) LSTM

Table 5: Hyperparameter Settings

For the weight matrices applied to the input and also the FC layers, we use Xavier uniform initialization, and since this network is not a ReLU type network (instead we're using tanh), we do not use Kaiming initialization. For weights applied to the hidden state, we use orthogonal initialization, which preserves gradient norm through each time step and alleviate vanishing/exploding gradient problem. We particularly select the bias for the forget gate to be 1, which is a standard Jozefowicz et al. (2015) recommendation, since initialization of values such as 0.5 will introduce vanishing gradient with a factor of 0.5 per timestep, therefore a forget gate bias of 1 encourages the LSTM to remember by default at the start of training, rather than being biased toward forgetting.

[4]The specific value is 0.00034561742212383306.

Bibliography

Baz, J., Granger, N., Harvey, C. R., Harvey, C. R., Roux, N. L., & Rattray, S. (2015). Dissecting investment strategies in the cross section and time series. https://api.semanticscholar.org/CorpusID:157964151

Jozefowicz, R., Zaremba, W., & Sutskever, I. (2015). An empirical exploration of recurrent network architectures. *Proceedings of the 32nd International Conference on Machine Learning - Volume 37*, 2342–2350.

Kryńska, K., & Ślepaczuk, R. (2022). *Daily and intraday application of various architectures of the lstm model in algorithmic investment strategies on bitcoin and the s&p 500 index* (Working Papers No. 2022-25). Faculty of Economic Sciences, University of Warsaw. https://ideas.repec.org/p/war/wpaper/2022-25.html

Tan, W. L., Roberts, S., & Zohren, S. (2024). Deep learning for options trading: An end-to-end approach. *Proceedings of the 5th ACM International Conference on AI in Finance*, 487–495. https://doi.org/10.1145/3677052.3698624